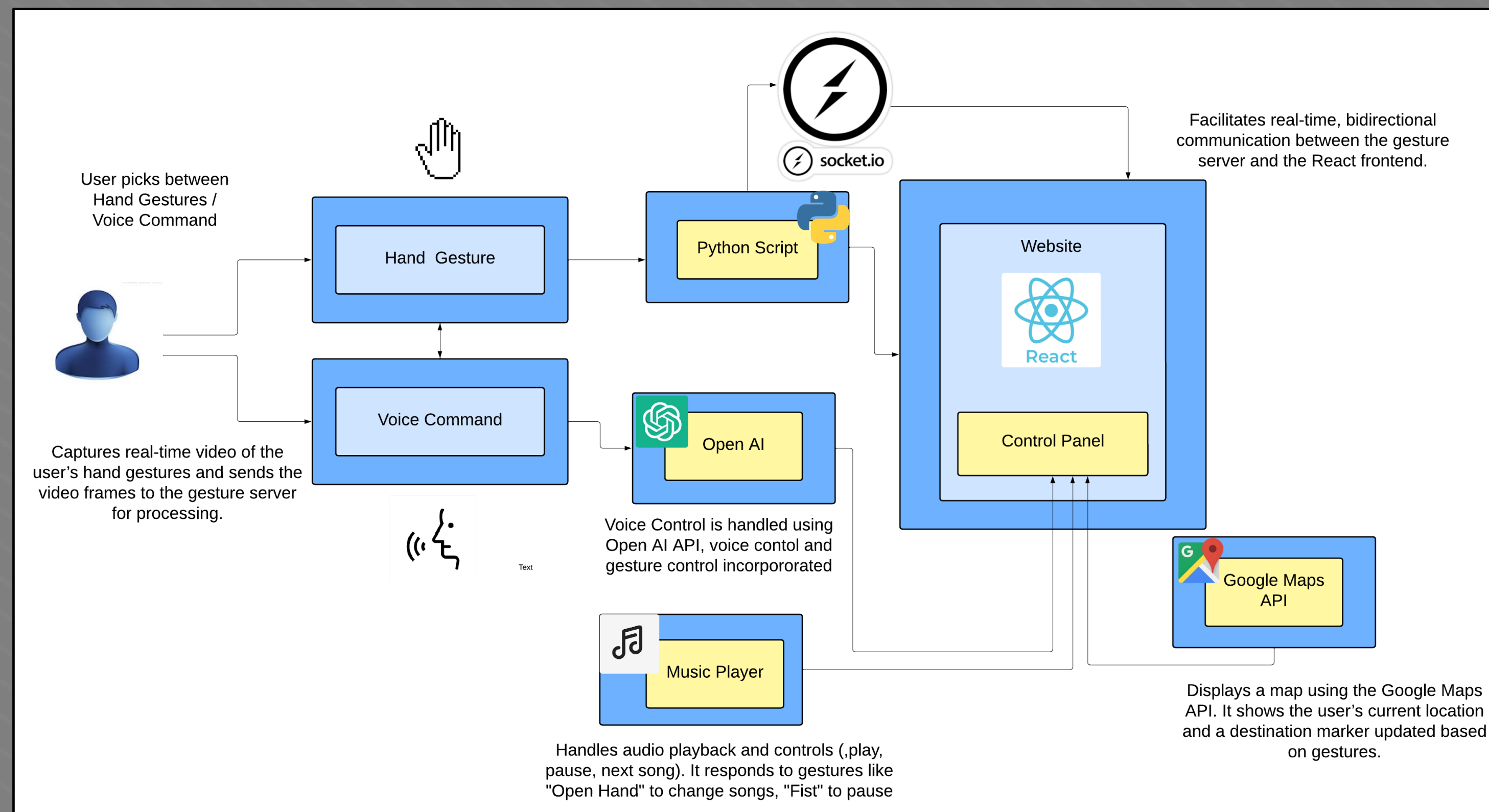


Summary

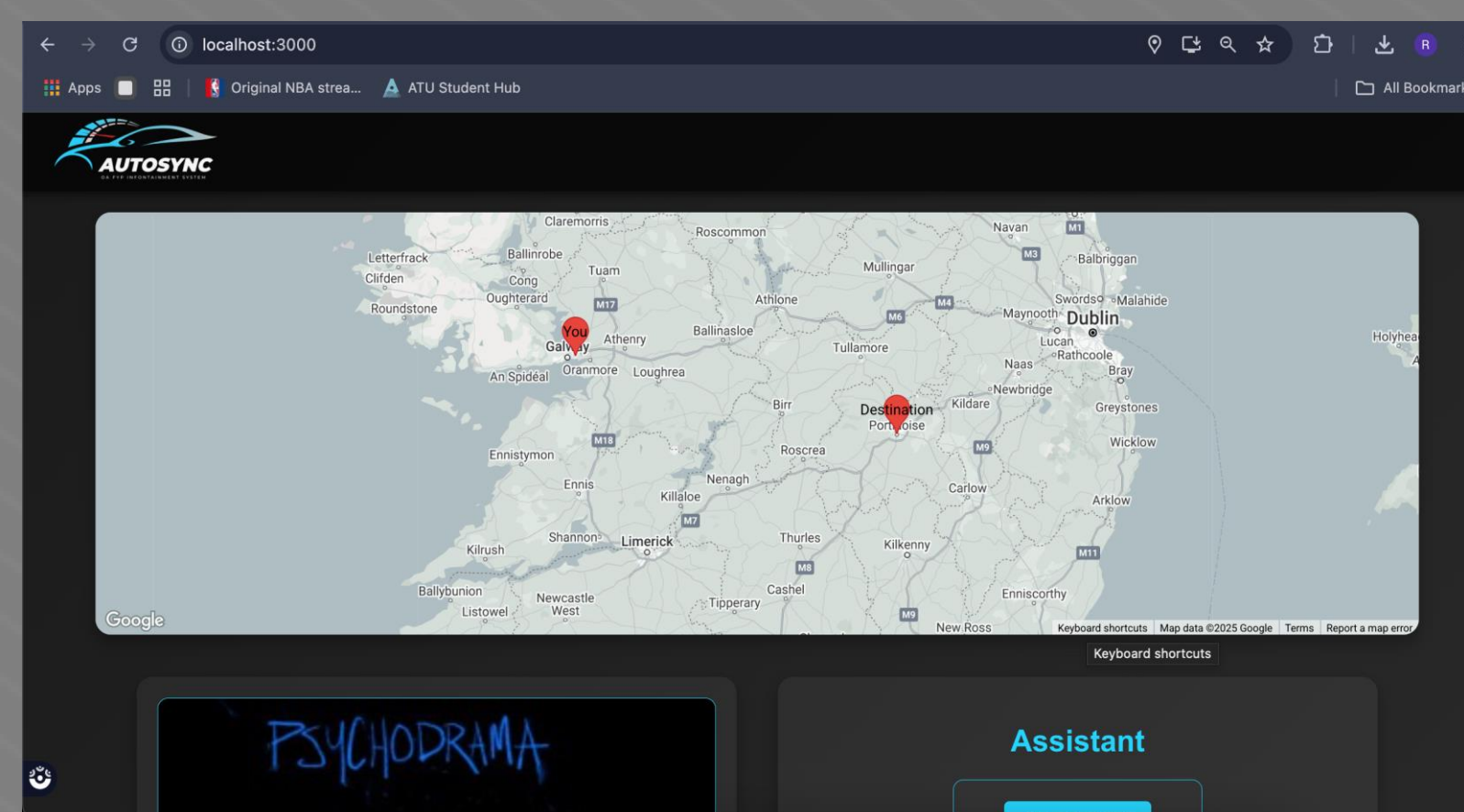
My infotainment system, allows users to control the navigation through Google Maps as well as the music player through defined hand gestures and voice commands.

The Gesture Server, written in Python using **MediaPipe** and **Socket.IO**, is responsible for detecting the gestures to set the car's destination as well as to make music selections, while the ChatGPT program processes the voice commands.

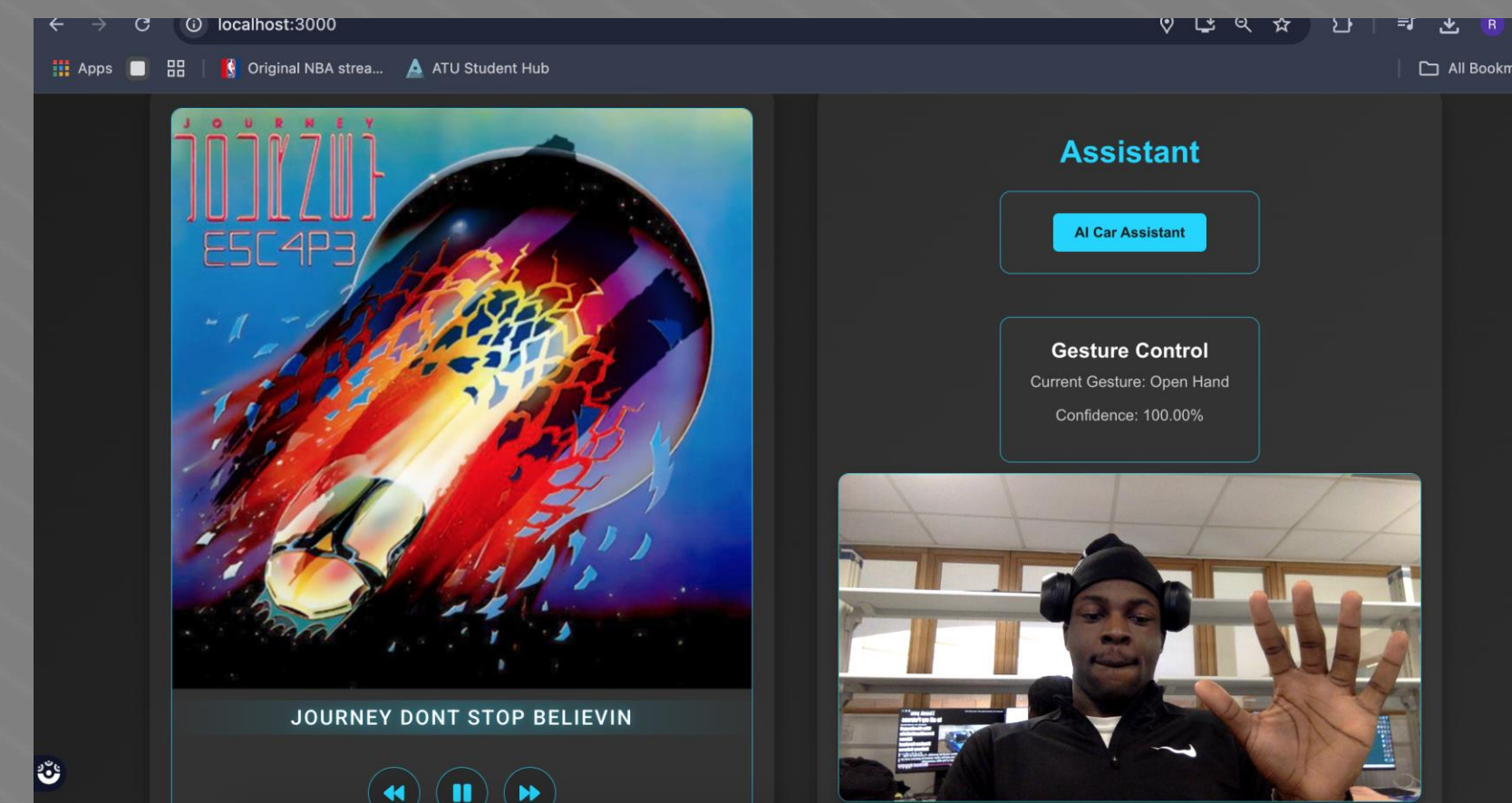
The React app ties everything together, changing modes between gesture-based and voice-based, and updates the map with Google Maps API, providing an enjoyable in car experience.



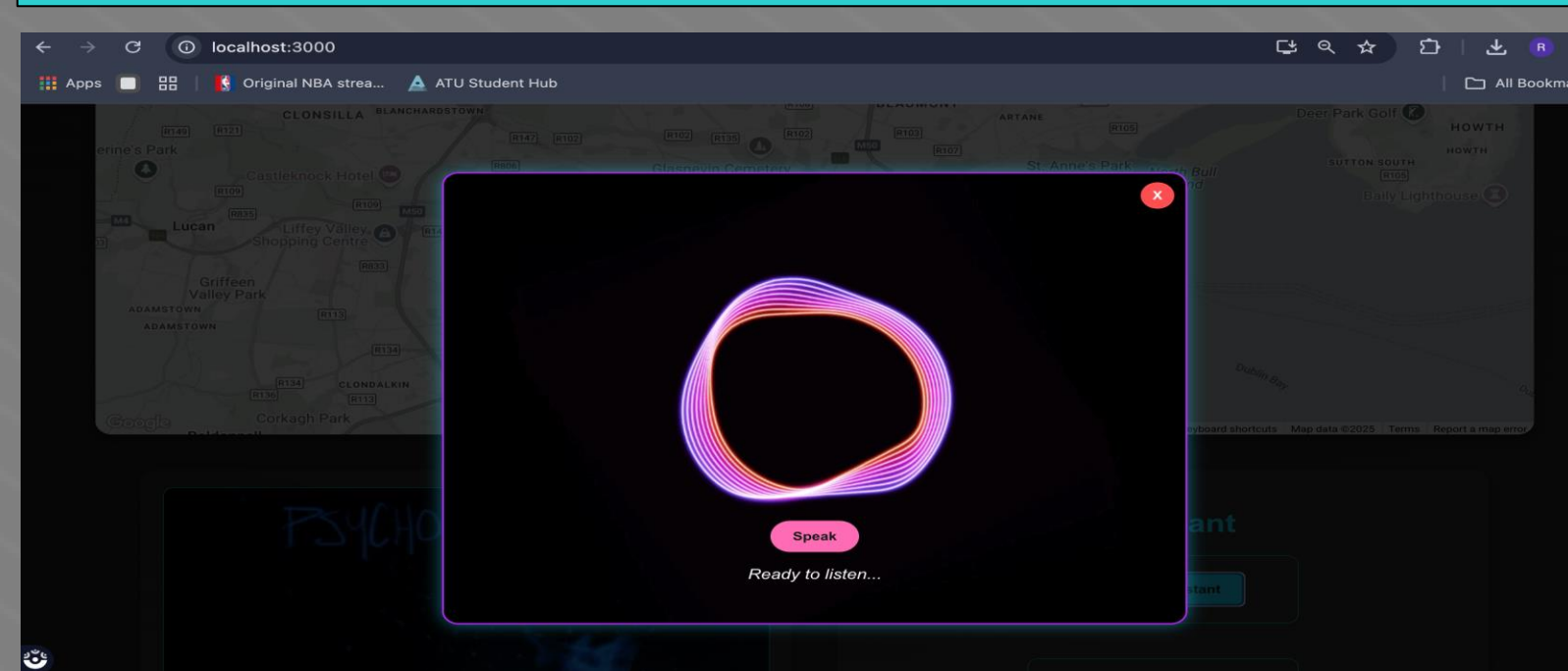
Features / Results



The **Google Maps API**, updates destination marker when triggered by a gesture or voice command.



The Music Player, enables song changes in the React App, responding to gesture detection.

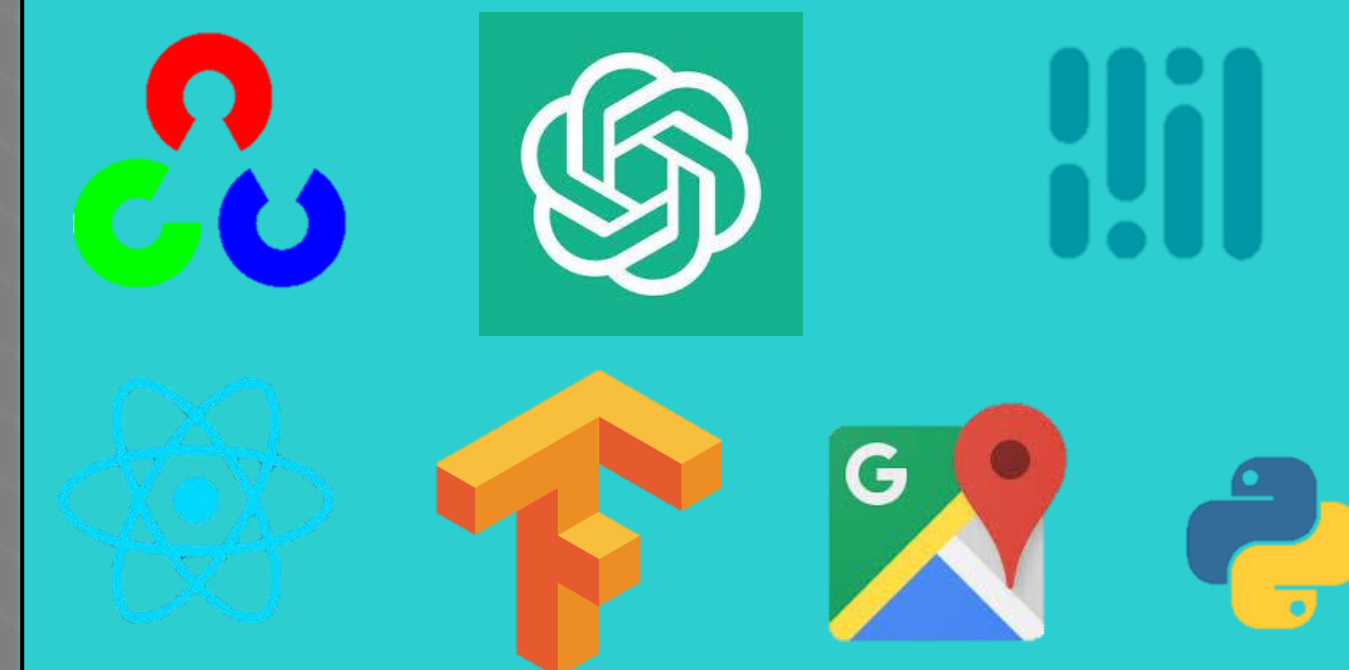


Voice commands are processed by **Open AI's ChatGPT**, allows users to set destinations, and ask questions.



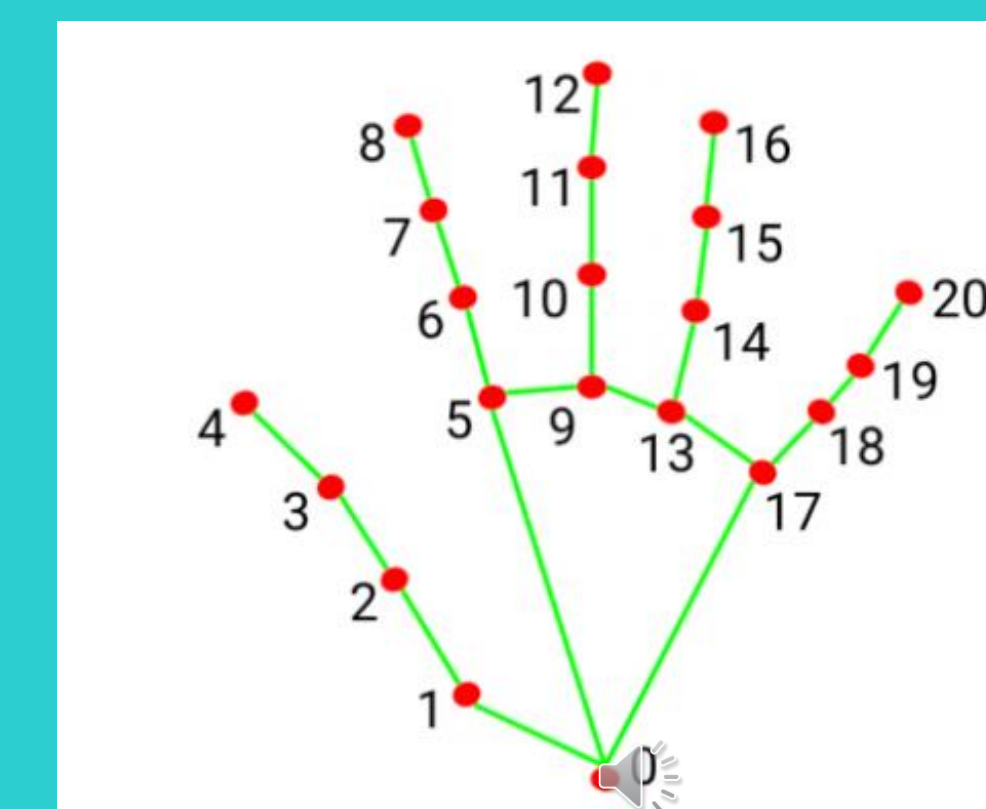
Results from hand gesture model using Mediapipe, assigning each gesture with a confidence score (up to 1.00) with smoothing.

Technology Used



- OpenCV**: Processes webcam for gesture detection.
- ChatGPT API**: Enables voice command AI interaction.
- MediaPipe**: Recognizes hand gestures for control.
- React**: Builds an interactive user interface.
- TensorFlow**: Platform used for gesture recognition model,.
- Google Maps API**: Displays maps and navigation markers.
- Python**: Runs gesture server and logic.

MediaPipe Gesture Recognition



MediaPipe Gesture Recognition is a Google machine learning-based system that recognizes and classifies hand gestures in real-time from a camera stream. It employs MediaPipe's hand tracking capability to find significant landmarks on the hand.

Inspiration

The motivation behind my car infotainment project was based on my interest in automotive tech, detection of hand gestures and the G-Series BMW design operating System 7.0 (MGU iDrive) bringing in intuitive gesture controls to operate features such as audio and calls with just hand gestures.

My project consists of a sophisticated infotainment system with sophisticated gesture recognition to facilitate safer, more engaging driving, inspired by the technological advancement and driver-focused innovation of BMW's G-Series cars.

